Do Macroeconomic and Geopolitical Risk Factors Improve FX Volatility Forecasts? A Mixed-Frequency Transformer Approach

Qimu Luo

Joint work with: Dr. Ren Zhao, Ruixi Xu, Zihan Liang, Zhuoqi Hou

International Business Economy Conference (IBEC)

January 4, 2026



Table of Contents

1 Introduction

2 Data Preparation

3 Methodology

4 Empirical Results
- Full Sample Analysis
- Regime-Specific Analysis

5 Case Study: 2024 U.S. Presidential Election Week

6 Conclusion

Motivation

Volatility Modeling:

- ★ Volatility forecasting is crucial for investment, option pricing, and financial regulation[1].
- ★ Reliable forecasts guide hedging, margin requirements, and central bank policy[2][3].
- ★ Application: Food, Climate, Commodity, Stocks, **Currency**

Map to FX Volatility:

- ★ FX markets are the largest and most liquid financial markets globally[4].
- ★ Property: Heteroskedasticity $\rightarrow$ volatility clustering $\rightarrow$ high persistency of **shocks**
- ★ Complexity: A complex interplay of **Market Sentiment** (High Frequency), **Economic Fundamentals** (Low Frequency), and **External Shocks** (Geopolitical Risk).

[1] Poon, S.-H. and Granger, C. W. J. (2003). Forecasting volatility in financial markets: A review. *Journal of economic literature*, 41(2), 478–539.

[2] Andersen, T. G. et al. (2003). Modeling and forecasting realized volatility. *Econometrica*, 71(2), 579–625.

[3] Christoffersen, P. and Diebold, F. X. (2012). Forecasting volatility in financial markets. *Handbook of economic forecasting*, 2, 453–497.

[4] Levinson, M. et al. (2014). *The economist guide to financial markets: Why they exist and how they work*. The Economist.

Evolution of Existing Models

Traditional Econometrics:

- ★ *ARCH*: Homoskedasticity → No persistence
- ★ *GARCH Family:* Linear or piecewise linear; Captures volatility clustering but struggles with non-linear, state-dependent interactions.
- ★ Limitation: Reply on historical time-series value, forecasting heavy, not the roles of economic fundamentals or GPR shocks

[5] Ersin, Ö. Ö. and Bildirici, M. (2023). Financial volatility modeling with the GARCH-MIDAS-LSTM approach: The effects of economic expectations, geopolitical risks and industrial production during COVID-19. *Mathematics*, 11(8), 1785.

[6] Naimy, V. et al. (2025). Redefining volatility forecasting in the aerospace and defense sector: application of CEEMDAN-GARCH models. *Humanities and Social Sciences Communications*, 12(1), 1–16.

[7] Fischer, T. et al. (2024). Fx-spot predictions with state-of-the-art transformer and time embeddings. *Expert Systems with Applications*, 249, 123538.

[8] Frank, J. (2023). *Forecasting realized volatility in turbulent times using temporal fusion transformers.* (FAU Discussion Papers in Economics No. [03/2023]). Retrieved from `https://ideas.repec.org/p/zbw/iwqwdp/032023.html`.

Evolution of Existing Models

Traditional Econometrics:

- ★ *ARCH*: Homoskedasticity → No persistence
- ★ *GARCH Family:* Linear or piecewise linear; Captures volatility clustering but struggles with non-linear, state-dependent interactions.
- ★ Limitation: Reply on historical time-series value, forecasting heavy, not the roles of economic fundamentals or GPR shocks

Deep Learning-based Approaches:

- ★ *Hybrid*: GARCH-MIDAS-LSTM capture nonlinear effects of macroeconomic variables[5]; Complementary Ensemble Empirical Mode Decomposition with Adaptive Noise + GARCH variants better capture volatility in aerospace[6].
- ★ Transformers: a Transformer with time embeddings[7], TFT (outperform GARCH)[8].
- ★ Limitation: Rely heavily on technical indicators, black-box models with limited economic insight.

[5] Ersin, Ö. Ö. and Bildirici, M. (2023). Financial volatility modeling with the GARCH-MIDAS-LSTM approach: The effects of economic expectations, geopolitical risks and industrial production during COVID-19. *Mathematics*, 11(8), 1785.

[6] Naimy, V. et al. (2025). Redefining volatility forecasting in the aerospace and defense sector: application of CEEMDAN-GARCH models. *Humanities and Social Sciences Communications*, 12(1), 1–16.

[7] Fischer, T. et al. (2024). Fx-spot predictions with state-of-the-art transformer and time embeddings. *Expert Systems with Applications*, 249, 123538.

[8] Frank, J. (2023). *Forecasting realized volatility in turbulent times using temporal fusion transformers.* (FAU Discussion Papers in Economics No. [03/2023]). Retrieved from `https://ideas.repec.org/p/zbw/iwqwdp/032023.html`.

Research Questions & Hypothesis

Main Research Question

Do macroeconomic fundamentals and geopolitical risk indicators improve USD/CNH exchange rate volatility forecasting beyond technical indicators?

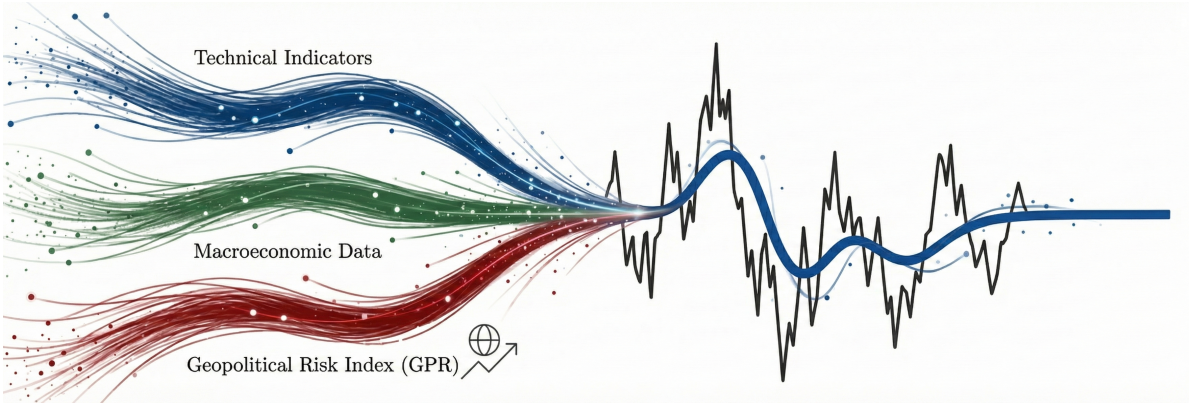
Research Questions & Hypothesis

Main Research Question

Do macroeconomic fundamentals and geopolitical risk indicators improve USD/CNH exchange rate volatility forecasting beyond technical indicators?

Research Hypothesis (H1): Incorporating macroeconomic fundamentals and geopolitical risk indicators improves USD/CNH volatility forecasts relative to a technical-only.

- ★ **H1a (Macroeconomic Channel – Expectation Anchoring):** Macroeconomic fundamentals enhance volatility forecasting by anchoring medium- to long-run exchange rate expectations.
- ★ **H1b (Geopolitical Risk Channel – Shock Amplification):** Geopolitical risk indicators improve short-horizon volatility forecasts by capturing uncertainty shocks and volatility spikes.
- ★ **H1c (Interaction Channel – Dynamic Modulation):** Jointly integrating macroeconomic and geopolitical information delivers complementary forecasting gains across different volatility regimes.

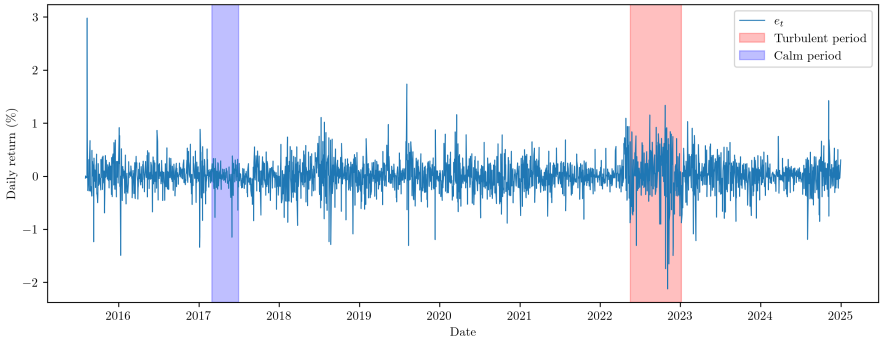


Method Contribution
Cross-attention fusion for heterogeneous data streams

Empirical Evaluation
Regime-dependent evidence across calm and stress periods

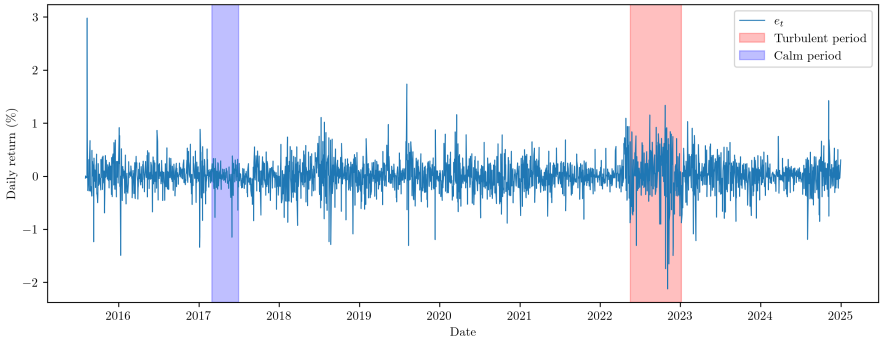
Risk Management Insights
Implications for event-sensitive currencies

Preliminary Data Check



USD/CNH Daily Return (Aug 2015–Dec 2024)

Preliminary Data Check



USD/CNH Daily Return (Aug 2015–Dec 2024)

- ★ Regime-specific behavior.
- ★ Periods of large fluctuations tend to persist, alternating with tranquil phases.

Feature Preparation: Technical Indicators

Technical Indicators serve as the base input for all streams models:

Table 1: USD/CNH Technical Indicators

Indicator	**Description**	**Computation**
SMA(20)	Simple Moving Average	Mean of close over the past 20 trading days.
EMA(50)	Exponential Moving Avg	EWMA of close with span 50.
RSI(14)	Relative Strength Index	Wilder's RSI, period 14; momentum oscillator.
ATR(14)	Average True Range	True range on OHLC; Wilder's smoothing.
RET	Daily Return	$r_t = \log(C_t/C_{t-1})$.
RET_MA(10)	Rolling Mean of Returns	10-day rolling mean of r_t.
RET_SD(10)	Rolling Std. of Returns	10-day rolling std. of r_t.

Feature Preparation: Macroeconomic Indicators

Selection Guidance: Classic Pricing Channels

- ★ **Purchasing Power Parity (PPP):** Relative prices and liquidity conditions.
- ★ **Interest Rate Parity (IRP):** Interest rate differentials and monetary policy conditions.
- ★ **Balance of Payments (BoP):** Trade and capital flow conditions.

[9] Baker, S. R. et al. (2016). Measuring economic policy uncertainty. *The quarterly journal of economics*, 131(4), 1593–1636.

Feature Preparation: Macroeconomic Indicators

Selection Guidance: Classic Pricing Channels

- ★ **Purchasing Power Parity (PPP):** Relative prices and liquidity conditions.
- ★ **Interest Rate Parity (IRP):** Interest rate differentials and monetary policy conditions.
- ★ **Balance of Payments (BoP):** Trade and capital flow conditions.

Table 2: USD/CNH Macroeconomic Indicators

Variable	Description	Details	Freq.	Source
CPI	Consumer Price Index	Headline and core CPI (CN, US)	Monthly	Wind
PPI	Producer Price Index	Producer prices (CN, US)	Monthly	Wind
MS (M2)	Money Supply	Broad money aggregates (CN, US)	Monthly	Wind
IR	Interest Rates	US Treasuries, SHIBOR/IBOR (CN)	Daily	Wind
FER	FX Reserves	Official reserves (CN, US)	Monthly	Wind; FRED
Employment	Employment Ratio	Employment (US)	Monthly	Wind
Gold	Gold Price	XAUUSD (US), SGE benchmark (CN)	Daily	Wind
Oil	Oil Price	WTI front-month benchmark	Daily	Wind
Stock	Equity Index	CSI 300 (CN), NASDAQ (US)	Daily	Wind
EPU	Policy Uncertainty	Global, CN, HK, US indices	Monthly	Policy Uncertainty[9]
EMV/VIX	Market Volatility	CBOE VIX; news-based EMV	Daily	Wind; CBOE

[9] Baker, S. R. et al. (2016). Measuring economic policy uncertainty. *The quarterly journal of economics*, 131(4), 1593–1636.

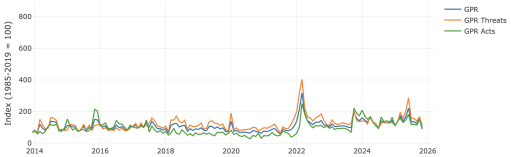
Feature Preparation: Geopolitical Risk Index

We proxy indirect economic shocks using the GPR index, which summarizes shocks to multiple moments of the geopolitical event distribution[10].

[10] Figlewski, S. (1997). Forecasting volatility. *Financial markets, institutions & instruments*, 6(1), 1–88.

Feature Preparation: Geopolitical Risk Index

We proxy indirect economic shocks using the GPR index, which summarizes shocks to multiple moments of the geopolitical event distribution[10].



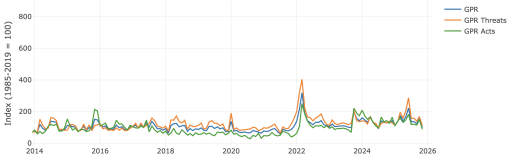
Global GPR, Threats, and Acts

- ★ GPR Threats: Reflects expectation; can move without realized events.
- ★ GPR Acts: Spikes align with realized geopolitical acts.

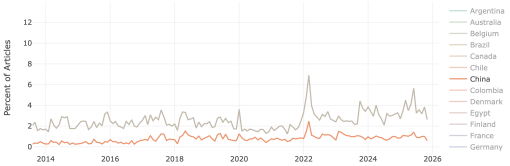
[10] Figlewski, S. (1997). Forecasting volatility. *Financial markets, institutions & instruments*, 6(1), 1–88.

Feature Preparation: Geopolitical Risk Index

We proxy indirect economic shocks using the GPR index, which summarizes shocks to multiple moments of the geopolitical event distribution[10].



Global GPR, Threats, and Acts



Country-Level GPR (CN vs. US)

- ★ GPR Threats: Reflects expectation; can move without realized events.
- ★ GPR Acts: Spikes align with realized geopolitical acts.

- ★ U.S. (grey): Dominant source of GPR coverage, driving major spikes.
- ★ China (orange): Smaller but rising share, co-moving with the U.S. during major spikes.

[10] Figlewski, S. (1997). Forecasting volatility. *Financial markets, institutions & instruments*, 6(1), 1–88.

Target Construction: Univariate Time-Series Forecasting

We forecast $\sigma^2_{t+h|t} \equiv \mathbb{E}\big(RV_{t,t+h} \mid \mathscr{F}_t\big)$, where $\mathscr{F}_t$ denotes the information set available at time t. Parkinson range-based daily variance[11] (a proxy for RV):

$$RV_t \approx \sigma^2_{pv,t} = \frac{1}{4\ln 2}\left(\ln \frac{H_t}{L_t}\right)^2.$$

and construct smoothed forecasting targets via

$$v_t^{\text{rolling}} = \sum_{i=0}^{w-1} \sigma^2_{pv,t-i}, \qquad v_t^{\text{target}} = \frac{1}{h}\sum_{i=1}^{h} v_{t+i},$$

where $w = 10$ denotes the rolling window and and $h \in \mathscr{H} = \{1, 5, 10\}$.

[11] Parkinson, M. (1980). The extreme value method for estimating the variance of the rate of return. *Journal of Business*, 53(1), 61–65.

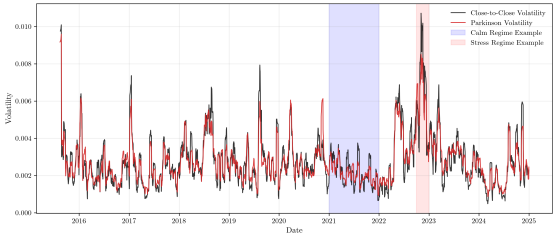
Why Parkinson Volatility?

Closing-price–based measures capture only a fraction of intraday dynamics, while range-based estimators outperform close-to-close volatility by filtering out distortions from bid–ask bounce and asynchronous trading[12].

[12] Alizadeh, S. et al. (2002). Range-based estimation of stochastic volatility models. *The Journal of Finance*, 57(3), 1047–1091.

Why Parkinson Volatility?

Closing-price–based measures capture only a fraction of intraday dynamics, while range-based estimators outperform close-to-close volatility by filtering out distortions from bid–ask bounce and asynchronous trading[12].

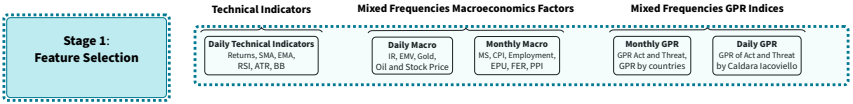


Close-to-Close (Black) vs. Parkinson Volatility (Red) at $w = 10$

- ★ Calm periods: Parkinson volatility is lower and smoother.
- ★ Stress periods: Both measures rise, but Parkinson avoids exaggerated spikes.

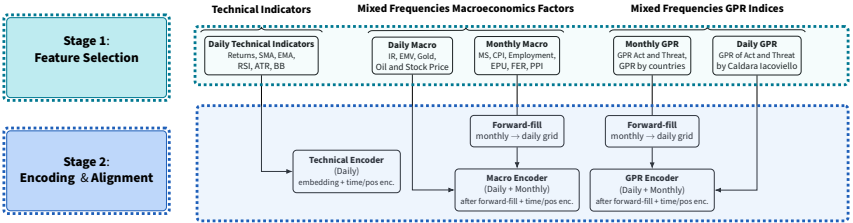
[12] Alizadeh, S. et al. (2002). Range-based estimation of stochastic volatility models. *The Journal of Finance*, 57(3), 1047–1091.

Whole Procedure: Mixed-Frequency Transformer



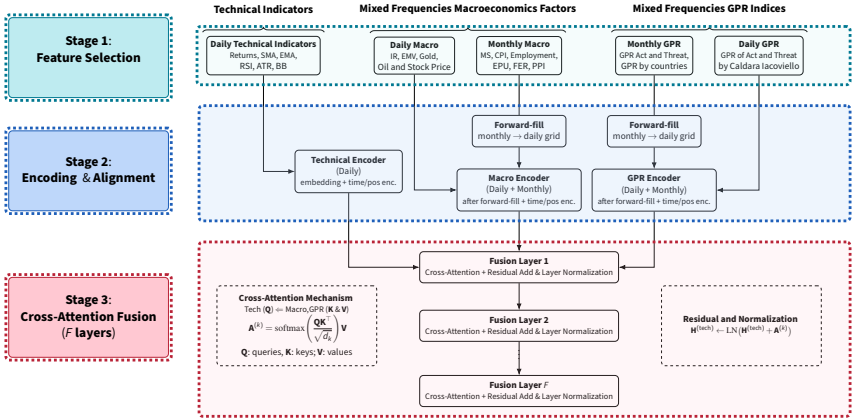
A Mixed-Frequency Transformer Procedure (Stage 1 – Stage 3)

Whole Procedure: Mixed-Frequency Transformer



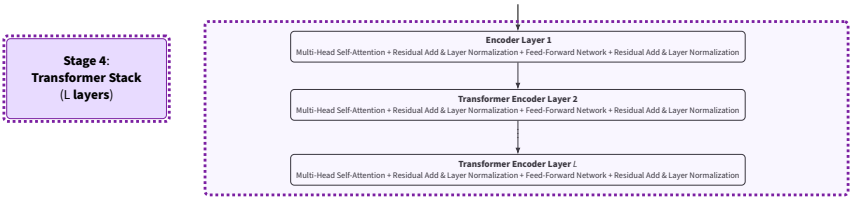
A Mixed-Frequency Transformer Procedure (Stage 1 – Stage 3)

Whole Procedure: Mixed-Frequency Transformer



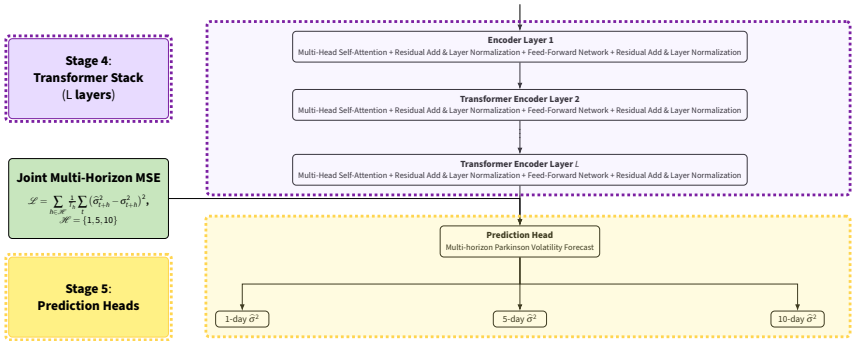
A Mixed-Frequency Transformer Procedure (Stage 1 – Stage 3)

Mixed-Frequency Transformer Procedure (Cont'd)



A Mixed-Frequency Transformer Procedure (Stage 4 – Stage 5)

Mixed-Frequency Transformer Procedure (Cont'd)



A Mixed-Frequency Transformer Procedure (Stage 4 – Stage 5)

Full-sample Performance: Comprehensive Evaluation

Table 3: Forecasting performance comparison across horizons.

Model	(w,h)	R^2 (%)	**RMSE** ($\times 10^{-4}$)	**MAE** ($\times 10^{-4}$)	**MAPE** (%)	**DA** (%)	**Spike-Hit** (%)
Transformer (Baseline)	Pooled	68.8	1.4	1.1	14.4	57.0	57.1
Macro-Augmented MFT	Pooled	61.7	1.6	1.3	19.7	58.1	53.6
GPR-Augmented MFT	Pooled	63.7	1.5	1.2	18.7	55.9	59.5
Joint Macro–GPR MFT	Pooled	44.9	1.8	1.5	25.9	**62.9**	**66.7**
Transformer (Baseline)	(10,1)	85.7	1.1	0.8	9.6	56.5	75.0
Macro-Augmented MFT	(10,1)	80.5	1.3	1.0	14.7	57.3	71.4
GPR-Augmented MFT	(10,1)	84.9	1.1	0.9	13.7	56.5	78.6
Joint Macro–GPR MFT	(10,1)	82.5	1.2	1.0	16.5	**62.9**	**89.3**
Transformer (Baseline)	(10,5)	73.7	1.4	1.0	14.0	60.5	60.7
Macro-Augmented MFT	(10,5)	69.8	1.5	1.2	17.7	60.5	50.0
GPR-Augmented MFT	(10,5)	69.8	1.5	1.2	18.1	59.4	64.3
Joint Macro–GPR MFT	(10,5)	65.0	1.6	1.4	22.8	**65.8**	**64.3**
Transformer (Baseline)	(10,10)	46.9	1.8	1.4	19.7	54.0	35.7
Macro-Augmented MFT	(10,10)	34.8	2.0	1.7	26.5	56.5	39.3
GPR-Augmented MFT	(10,10)	36.5	2.0	1.6	24.3	51.8	35.7
Joint Macro–GPR MFT	(10,10)	-13.0	2.7	2.2	38.4	**60.0**	**46.4**

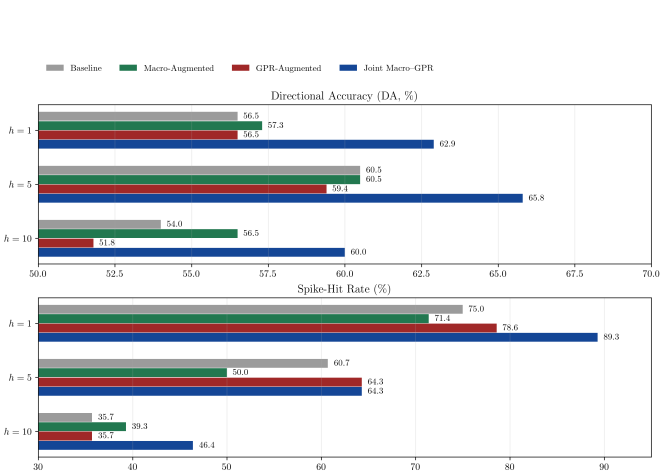
Full-sample Performance: Comprehensive Evaluation

Table 3: Forecasting performance comparison across horizons.

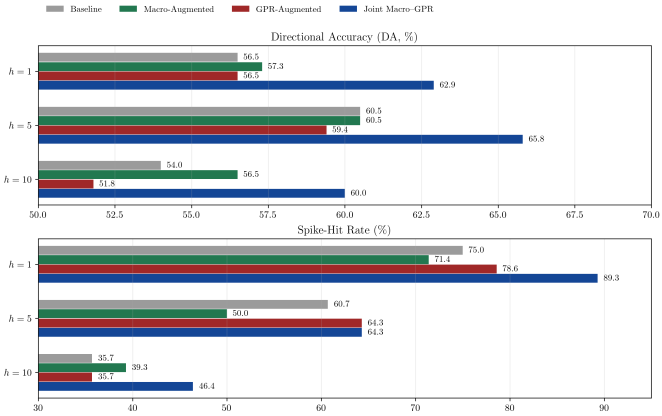
Model	(w,h)	R^2 (%)	**RMSE** ($\times 10^{-4}$)	**MAE** ($\times 10^{-4}$)	**MAPE** (%)	**DA** (%)	**Spike-Hit** (%)
Transformer (Baseline)	Pooled	68.8	1.4	1.1	14.4	57.0	57.1
Macro-Augmented MFT	Pooled	61.7	1.6	1.3	19.7	58.1	53.6
GPR-Augmented MFT	Pooled	63.7	1.5	1.2	18.7	55.9	59.5
Joint Macro–GPR MFT	Pooled	44.9	1.8	1.5	25.9	**62.9**	**66.7**
Transformer (Baseline)	(10,1)	85.7	1.1	0.8	9.6	56.5	75.0
Macro-Augmented MFT	(10,1)	80.5	1.3	1.0	14.7	57.3	71.4
GPR-Augmented MFT	(10,1)	84.9	1.1	0.9	13.7	56.5	78.6
Joint Macro–GPR MFT	(10,1)	82.5	1.2	1.0	16.5	**62.9**	**89.3**
Transformer (Baseline)	(10,5)	73.7	1.4	1.0	14.0	60.5	60.7
Macro-Augmented MFT	(10,5)	69.8	1.5	1.2	17.7	60.5	50.0
GPR-Augmented MFT	(10,5)	69.8	1.5	1.2	18.1	59.4	64.3
Joint Macro–GPR MFT	(10,5)	65.0	1.6	1.4	22.8	**65.8**	**64.3**
Transformer (Baseline)	(10,10)	46.9	1.8	1.4	19.7	54.0	35.7
Macro-Augmented MFT	(10,10)	34.8	2.0	1.7	26.5	56.5	39.3
GPR-Augmented MFT	(10,10)	36.5	2.0	1.6	24.3	51.8	35.7
Joint Macro–GPR MFT	(10,10)	-13.0	2.7	2.2	38.4	**60.0**	**46.4**

- ★ Trade-off: Higher risk sensitivity comes at the cost of point accuracy.
- ★ Horizon & roles: GPR matters short-run, macro fundamentals dominate longer horizons.
- ★ Implication: Full-sample results motivate regime-specific analysis.

Full-sample Performance: Risk-Oriented Insights



Full-sample Performance: Risk-Oriented Insights



- ★ **Joint model dominance**: Across all horizons, the Joint Macro–GPR model consistently achieves the highest Directional Accuracy and Spike-Hit rate, indicating superior risk sensitivity.
- ★ **Horizon asymmetry**: Gains are strongest at short horizons ($h = 1$), especially for spike detection, and attenuate as horizons lengthen.

Regime Analysis: Defining the Market State

Volatility clustering $\rightarrow$ How does each model's role differ in various states?

We differentiate between two market regimes:

Stress Regime

- ★ **Definition:** Top 20% of volatility months.
- ★ **Period:** Aug, Sep, Nov 2024.
- ★ **Characteristic:** High uncertainty, frequent spikes.

Calm Regime

- ★ **Definition:** Bottom 40% of volatility months in the test set.
- ★ **Period:** March to July 2024.
- ★ **Characteristic:** Stable trends, low noise.

Stress Regime: Comprehensive Evaluation

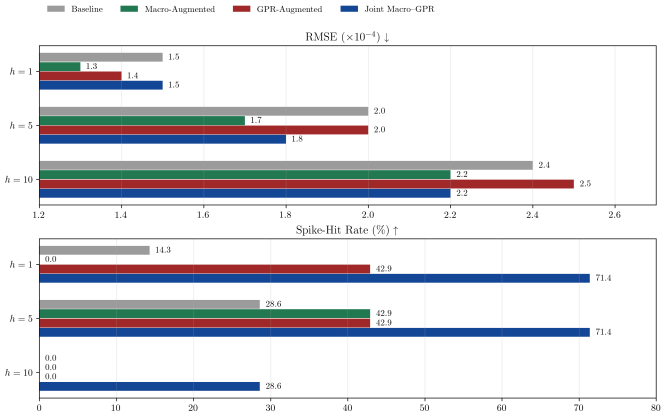
Table 4: Forecasting performance in the stress regime.

Model	(w,h)	**RMSE** ($\times 10^{-4}$)	**MAE** ($\times 10^{-4}$)	**MAPE** (%)	**DA** (%)	**Spike-Hit** (%)
Transformer (Baseline)	(10,1)	1.5	1.0	9.4	58.7	14.3
Macro-Augmented MFT	(10,1)	1.3	0.8	7.7	63.5	0.0
GPR-Augmented MFT	(10,1)	1.4	1.1	11.4	61.9	42.9
Joint Macro–GPR MFT	(10,1)	1.5	1.2	13.7	**74.6**	**71.4**
Transformer (Baseline)	(10,5)	2.0	1.5	14.4	61.9	28.6
Macro-Augmented MFT	(10,5)	1.7	1.1	10.3	61.9	42.9
GPR-Augmented MFT	(10,5)	2.0	1.6	16.5	66.7	42.9
Joint Macro–GPR MFT	(10,5)	1.8	1.4	14.9	**69.8**	**71.4**
Transformer (Baseline)	(10,10)	2.4	1.8	18.7	50.8	0.0
Macro-Augmented MFT	(10,10)	2.2	1.7	17.2	55.6	0.0
GPR-Augmented MFT	(10,10)	2.5	1.9	19.9	52.4	0.0
Joint Macro–GPR MFT	(10,10)	2.2	1.9	20.3	**54.0**	**28.6**

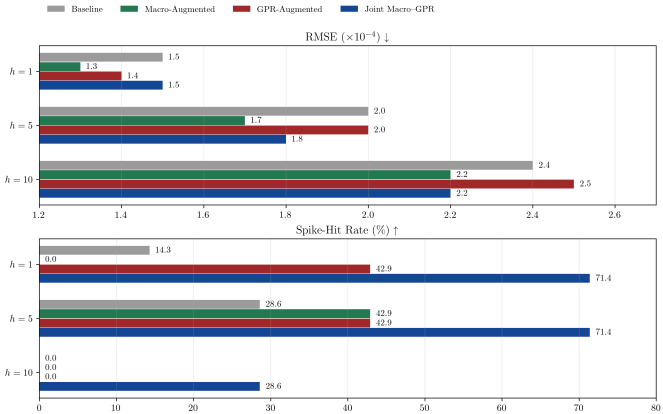
★ The Joint Macro–GPR MFT: Retains the risk sensitivity dominates in stress; Faces tradeoff in terms of point accuracy (RMSE/MAE/MAPE)

★ Distinct roles of macro vs. GPR

Stress Regime: Macro- vs. GPR-Augmented MFT



Stress Regime: Macro- vs. GPR-Augmented MFT



★ **Point-Accuracy**: Across all horizons, the Macro–Augmented MFT consistently achieves the lowest RMSE → highest accuracy

★ **Risk Sensitivity**: GPR-Augmented MFT improves Spike-Hit rate at short and medium horizons

Calm Regime: Comprehensive Evaluation

Table 5: Forecasting performance in the calm regime.

Model	(w,h)	**RMSE** ($\times 10^{-4}$)	**MAE** ($\times 10^{-4}$)	**MAPE** (%)	**DA** (%)	**Spike-Hit** (%)
Transformer (Baseline)	(10,1)	0.8	0.6	9.8	52.4	0.0
Macro-Augmented MFT	(10,1)	1.0	0.9	20.7	53.4	0.0
GPR-Augmented MFT	(10,1)	0.9	0.8	18.5	53.4	0.0
Joint Macro–GPR MFT	(10,1)	1.1	1.0	24.0	**62.1**	**63.6**
Transformer (Baseline)	(10,5)	1.3	0.9	16.9	62.1	9.1
Macro-Augmented MFT	(10,5)	1.5	1.3	26.8	64.1	9.1
GPR-Augmented MFT	(10,5)	1.5	1.2	25.9	62.1	9.1
Joint Macro–GPR MFT	(10,5)	1.9	1.8	39.7	**68.9**	**90.9**
Transformer (Baseline)	(10,10)	1.8	1.4	27.3	64.1	18.2
Macro-Augmented MFT	(10,10)	2.4	2.2	46.7	59.2	72.7
GPR-Augmented MFT	(10,10)	2.2	1.9	38.7	57.3	36.4
Joint Macro–GPR MFT	(10,10)	3.8	3.6	77.1	**68.0**	**100.0**

Calm Regime: Comprehensive Evaluation

Table 5: Forecasting performance in the calm regime.

Model	(w,h)	**RMSE** ($\times 10^{-4}$)	**MAE** ($\times 10^{-4}$)	**MAPE** (%)	**DA** (%)	**Spike-Hit** (%)
Transformer (Baseline)	(10,1)	0.8	0.6	9.8	52.4	0.0
Macro-Augmented MFT	(10,1)	1.0	0.9	20.7	53.4	0.0
GPR-Augmented MFT	(10,1)	0.9	0.8	18.5	53.4	0.0
Joint Macro–GPR MFT	(10,1)	1.1	1.0	24.0	**62.1**	**63.6**
Transformer (Baseline)	(10,5)	1.3	0.9	16.9	62.1	9.1
Macro-Augmented MFT	(10,5)	1.5	1.3	26.8	64.1	9.1
GPR-Augmented MFT	(10,5)	1.5	1.2	25.9	62.1	9.1
Joint Macro–GPR MFT	(10,5)	1.9	1.8	39.7	**68.9**	**90.9**
Transformer (Baseline)	(10,10)	1.8	1.4	27.3	64.1	18.2
Macro-Augmented MFT	(10,10)	2.4	2.2	46.7	59.2	72.7
GPR-Augmented MFT	(10,10)	2.2	1.9	38.7	57.3	36.4
Joint Macro–GPR MFT	(10,10)	3.8	3.6	77.1	**68.0**	**100.0**

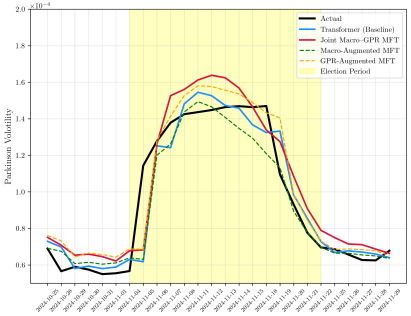
★ Point forecasting: The baseline Transformer achieves the best point accuracy, indicating that technical signals are sufficient when volatility is low.

★ Risk monitoring: The Joint Macro–GPR MFT delivers the highest directional accuracy and spike detection, revealing latent risks even in tranquil markets.

Case Study: Results

Test the robustness of our model under real political stress:

Model	R^2 (%)	**RMSE** ($\times 10^{-4}$)	**MAE** ($\times 10^{-4}$)	**DA (%)**
Transformer	57.6	1.7	1.1	53.8
Macro-Augmented MFT	58.6	1.7	1.1	69.2
GPR-Augmented MFT	**61.2**	**1.6**	1.1	69.2
Joint Macro–GPR MFT	53.4	1.8	1.5	**76.9**

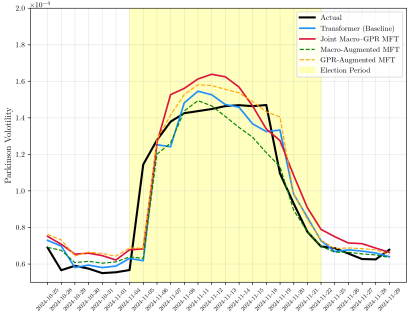


2024 U.S. Presidential Election Window (Nov 5–Nov 22)

Case Study: Results

Test the robustness of our model under real political stress:

Model	R^2 (%)	**RMSE** ($\times 10^{-4}$)	**MAE** ($\times 10^{-4}$)	**DA (%)**
Transformer	57.6	1.7	1.1	53.8
Macro-Augmented MFT	58.6	1.7	1.1	69.2
GPR-Augmented MFT	**61.2**	**1.6**	1.1	69.2
Joint Macro–GPR MFT	53.4	1.8	1.5	**76.9**



2024 U.S. Presidential Election Window (Nov 5–Nov 22)

★ GPR is the key driver of shock sensitivity: GPR-only best captures the election volatility spike (highest R^2 at 61.2%, lowest RMSE).

★ Macro and GPR inputs sharply improve stress detection: the highest DA (76.9%) despite lower point-forecast accuracy.

Case Study: Feature Attribution

To assess model interpretability, we adopt the Integrated Gradients (IG) method[13]:

$$IG_i(X;X') = (x_i - x_i') \int_0^1 \frac{\partial f\big(x' + \alpha(x - x')\big)}{\partial x_i}\, d\alpha,$$

where X' and X denote the baseline and actual inputs.

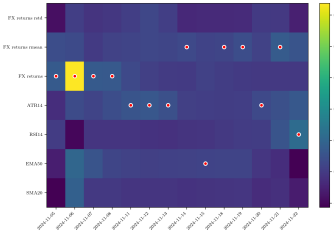
[13] Sundararajan, M. et al. (June 2017). Axiomatic Attribution for Deep Networks. In Precup, D. and Teh, Y. W.*Proceedings of the 34th International Conference on Machine Learning* (pp. 3319–3328). PMLR..

Case Study: Feature Attribution

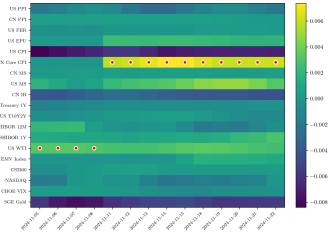
To assess model interpretability, we adopt the Integrated Gradients (IG) method[13]:

$$IG_i(X;X') = (x_i - x_i') \int_0^1 \frac{\partial f\big(x' + \alpha(x - x')\big)}{\partial x_i}\, d\alpha,$$

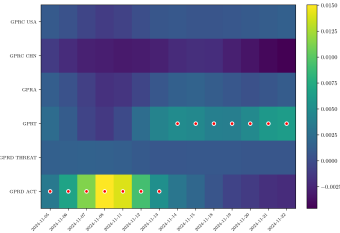
where X' and X denote the baseline and actual inputs.



Technical Indicators Attribution at $h = 1$



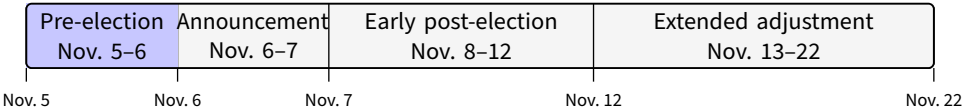
Macro Indicators Attribution at $h = 1$



GPR Indicators Attribution at $h = 1$

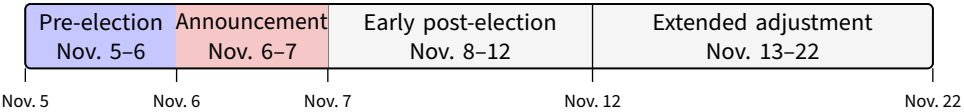
[13] Sundararajan, M. et al. (June 2017). Axiomatic Attribution for Deep Networks. In Precup, D. and Teh, Y. W.*Proceedings of the 34th International Conference on Machine Learning* (pp. 3319–3328). PMLR..

Election Phases Timeline



Pre-election uncertainty: FX returns, realized volatility (short-term momentum).

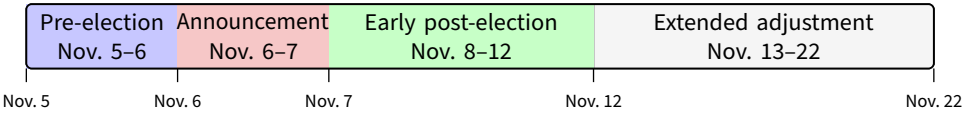
Election Phases Timeline



Pre-election uncertainty: FX returns, realized volatility (short-term momentum).

Announcement shock: GPR Acts (daily), GPR Threats.

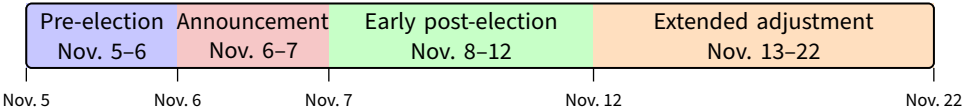
Election Phases Timeline



Pre-election uncertainty: FX returns, realized volatility (short-term momentum).

Announcement shock: GPR Acts (daily), GPR Threats. **Early post-election adjustment:** CPI (inflation expectations), oil prices (WTI), money supply.

Election Phases Timeline



Pre-election uncertainty: FX returns, realized volatility (short-term momentum).

Announcement shock: GPR Acts (daily), GPR Threats. **Early post-election adjustment:** CPI (inflation expectations), oil prices (WTI), money supply.

Extended adjustment: Technical trend indicators (ATR, RSI, EMA), reduced GPR influence.

Insights for Different Stakeholders

For Risk Managers

- ★ **Horizon-specific:** technical models suffice in tranquil markets.
- ★ **Crisis monitoring:** GPR captures event-driven volatility spikes.
- ★ **Tail-risk alerts:** Joint MFT excels in spike-hit detection.

Insights for Different Stakeholders

For Risk Managers

- ★ **Horizon-specific:** technical models suffice in tranquil markets.
- ★ **Crisis monitoring:** GPR captures event-driven volatility spikes.
- ★ **Tail-risk alerts:** Joint MFT excels in spike-hit detection.

For Quant Investors

- ★ **Event-driven alpha:** combining macro and GPR improves timing of directional shifts and spikes.
- ★ Use baseline models for levels and Joint MFT for regime-aware hedging.

Insights for Different Stakeholders

For Risk Managers

- ★ **Horizon-specific:** technical models suffice in tranquil markets.
- ★ **Crisis monitoring:** GPR captures event-driven volatility spikes.
- ★ **Tail-risk alerts:** Joint MFT excels in spike-hit detection.

For Quant Investors

- ★ **Event-driven alpha:** combining macro and GPR improves timing of directional shifts and spikes.
- ★ Use baseline models for levels and Joint MFT for regime-aware hedging.

For Policymakers

- ★ **Asymmetric drivers**: fundamentals vs. political shocks.
- ★ **U.S. macro news** shifts Fed expectations and transmits globally.
- ★ **Geopolitical shocks** cause short-lived bursts; energy fundamentals and liquidity management mitigate depreciation pressure.

Conclusion

- ★ **What We Do**: We assess whether macroeconomic fundamentals and geopolitical risk indicators improve USD/CNH volatility forecasting within a mixed-frequency Transformer framework.

Conclusion

- ★ **What We Do**: We assess whether macroeconomic fundamentals and geopolitical risk indicators improve USD/CNH volatility forecasting within a mixed-frequency Transformer framework.
- ★ **Methodological Innovation**: We develop a **Mixed-Frequency Transformer** that jointly learns from daily market data, monthly macroeconomic variables, and high-frequency geopolitical risk signals without manual aggregation.

Conclusion

- ★ **What We Do**: We assess whether macroeconomic fundamentals and geopolitical risk indicators improve USD/CNH volatility forecasting within a mixed-frequency Transformer framework.
- ★ **Methodological Innovation**: We develop a **Mixed-Frequency Transformer** that jointly learns from daily market data, monthly macroeconomic variables, and high-frequency geopolitical risk signals without manual aggregation.
- ★ **Validation of Model Hypotheses**:
  - ☆ H1 (Overall Effect): Supported. Macro- and GPR-augmented models outperform the technical benchmark in **risk-sensitive metrics**, with limited gains in point-error measures.
  - ☆ H1a (Macroeconomic Channel): Partially supported. Macroeconomic fundamentals improve forecasts mainly at **medium–long horizons** and in **calm regimes**, with little impact on short-horizon spikes.
  - ☆ H1b (Geopolitical Risk Channel): Supported. GPR indicators significantly enhance **short-horizon spike detection** and **stress-regime performance**.
  - ☆ H1c (Interaction Channel): Partially supported. Joint macro–GPR models deliver **complementary, regime-dependent gains**.

Thank you!